

REGULATORY INTELLIGENCE

Domain Signal Extraction for Supply Chain Optimization

Technical README — Weekly signal pipeline, data dictionary, and integration guide.

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1. Overview

This pipeline transforms raw regulatory updates from 600+ regulators into weekly numeric signals that can feed directly into supply chain LP/MILP models as external context variables.

The core idea: regulatory data does not enter the LP as a decision variable. It conditions the LP's parameters — cost coefficients, capacity bounds, lead times, and corridor eligibility — so the model reflects current regulatory reality rather than stale assumptions.

1.1 Pipeline stages

1. **Scan** — Read annotation batch files (~100K artifacts, ~600 regulators)
2. **Score** — Rank regulators by frequency (data), geo fit and client fit (LLM)
3. **Extract signals** — Classify each annotation by domain and corridor, aggregate weekly
4. **Map to parameters** — Threshold-triggered rules adjust LP coefficients
5. **Solve & compare** — Run LP under baseline, stress, and relaxed scenarios

2. Signal domains

Each annotation is classified into one or more of eight regulatory domains. Each domain maps to specific LP parameter families.

Domain	Description	LP parameters affected
trade_customs	Trade, tariffs, customs, export controls, sanctions	cost_import, lead_time, corridor_eligibility
aml_kyc	Anti-money laundering, KYC, financial crime compliance	compliance_staff, onboarding_cost, due_diligence_days
fx_capital	Foreign exchange controls, capital movement, cross-border payments	txn_cost, payment_route_eligible, fx_premium
product_safety	Product standards, food safety, health regulations	capacity_port, inspection_days, certification_cost
cyber_data	Cybersecurity mandates, data protection, data localization	it_cost, digital_corridor_eligible, data_transfer_cost
labor_social	Labor law, working conditions, minimum wage, local content	workforce_cost, operating_hours, local_content_pct
inventory_warehose use	Storage regulations, bonded zones, hazmat, zoning	holding_cost, storage_capacity, bonded_zone_eligible
environmental_esg	ESG due diligence, environmental permits, carbon, CSDDD	corridor_eligibility, compliance_cost, carbon_surcharge

Annotations can match multiple domains. For example, CSDDD (Corporate Sustainability Due Diligence Directive) updates classify as both *trade_customs* and *environmental_esg*, because they affect both import costs and compliance obligations.

3. Geographic corridors

Each annotation is assigned to one or more geographic corridors based on the regulator's base URL domain, the annotation's tags, and language. This ensures that the signal reflects where the regulatory pressure actually applies, not just where the regulator is based.

Corridor	Matching signals	Example regulators
france_domestic	.gouv.fr, .fr domains, tags containing 'France'	AMF, ANSSI, CNIL, DGFiP
eu	europa.eu, EUR-Lex, EU-level bodies	European Commission, EBA, ESMA
french_africa	BCEAO, UEMOA, .sn/.ci/.ml domains, CFA-related tags	BCEAO, BRVM, national central banks
maghreb	.ma/.dz/.tn domains, Maghreb/North Africa tags	Bank Al-Maghrib, AMMC
gcc	.ae/.sa/.qa domains, Gulf/GCC tags	UAE Central Bank, SAMA, QFC
mena_other	.eg/.lb/.jo domains, MENA tags	CBE, BDL, CBJ
global	Fallback when no specific geography detected	FATF, BIS, ISO

4. Signal computation

4.1 Weekly aggregation

For each (domain, corridor) pair, annotations are grouped by ISO week. The raw signal for a given week is:

$$\text{signal} = \sum (\text{score_impact}_i \times \text{urgency_weight}_i)$$

$$\text{urgency_weight} = 1.0 + \text{score_urgency} / 10$$

This means a high-impact, high-urgency update (impact=9, urgency=8) contributes $9 \times 1.8 = 16.2$ to the weekly signal, while a high-impact but non-urgent update (impact=9, urgency=0) contributes 9.0. Both matter, but time-sensitive changes get amplified.

4.2 Trigger mechanism

The LP parameter adjustment fires when sustained regulatory pressure exceeds historical norms:

- **4-week rolling mean** — smooths out single-week spikes while catching sustained acceleration within one month.
- **80th percentile baseline** — computed over the trailing 52-week history for the same domain-corridor pair.

- **Trigger condition** — when $\text{rolling_4w_mean} > \text{baseline_p80}$, the trigger is active and the corresponding LP parameters should be adjusted.

The trigger is not binary in practice — the magnitude of the overshoot ($\text{rolling_4w_mean} / \text{baseline_p80}$) can scale the parameter adjustment proportionally.

5. Output data dictionary

The signals CSV contains one row per (week, domain, corridor) combination.

Column	Type	Description
week	string	ISO week identifier, e.g. 2025-W47
week_start	date	Monday of the ISO week (YYYY-MM-DD)
domain	string	One of the 8 signal domain names
corridor	string	Geographic corridor label
volume	int	Count of classified annotation-events in this week-domain-corridor
sum_impact	float	Sum of score_impact values across events
avg_impact	float	Mean score_impact (severity indicator)
max_urgency	int	Maximum score_urgency in the week (alert flag)
sum_weighted	float	Sum of (impact × urgency_weight) — same as signal
signal	float	The composite weekly signal value (primary metric)
rolling_4w_mean	float	4-week rolling mean of signal (smoothed trend)
baseline_p80	float	80th percentile of signal over trailing 52 weeks
trigger_active	bool	True when $\text{rolling_4w_mean} > \text{baseline_p80}$

6. Sample data profile

From the current extraction (30 top regulators, 12-month lookback):

- 1,066 weekly signal rows across 63 weeks
- 8 active domains + 1 unclassified residual
- 7 geographic corridors (france_domestic, eu, french_africa, maghreb, gcc, mena_other, global)
- 375 active trigger events (rolling 4-week mean exceeding 80th percentile baseline)

Highest-volume domains: aml_kyc (3,748 events, mean signal 138.9) and trade_customs (1,605 events, mean signal 91.2). These two domains dominate because the top-ranked regulators for this client profile are concentrated in financial regulation and EU trade bodies.

Highest-pressure corridors: gcc (mean signal 107.6), france_domestic (99.2), and eu (80.1). The GCC corridor's high signal reflects concentrated activity from a smaller number of regulators publishing high-impact updates.

7. Parameter mapping reference

When a trigger is active for a given domain-corridor, the following LP parameters should be adjusted. The multipliers below are starting points — they should be calibrated against historical disruption events for each client's specific supply chain.

Domain	Parameter adjustment	Suggested default
trade_customs	cost_import[corridor] × multiplier	1.10 – 1.20
trade_customs	lead_time[corridor] × multiplier	1.15 – 1.30
aml_kyc	compliance_staff[jurisdiction] += N	+1 to +3 FTE
aml_kyc	onboarding_cost[route] × multiplier	1.05 – 1.15
fx_capital	txn_cost[payment_route] × multiplier	1.05 – 1.12
fx_capital	payment_route_eligible = false	Binary toggle
product_safety	capacity[port] × multiplier	0.80 – 0.90
product_safety	inspection_days[origin] += N	+2 to +5 days
cyber_data	it_cost[node] × multiplier	1.08 – 1.15
cyber_data	digital_corridor_eligible = false	Binary toggle
labor_social	workforce_cost[location] × multiplier	1.05 – 1.15
inventory_warehouse	holding_cost[warehouse] × multiplier	1.05 – 1.10
environmental_esg	corridor_eligibility = false	Binary toggle
environmental_esg	carbon_surcharge[corridor] += rate	Per-unit cost

8. Usage

8.1 CLI commands

Step 1: Score regulators

```
python process.py score-regulators \
  --annotations-dir data/annotations \
  --region "France, French-speaking Africa, MENA" \
  --client "Supply chain consultancy, France/Africa/MENA" \
  --output scored_regulators.csv
```

Step 2: Extract annotations for top regulators

```
python process.py extract-annotations \
  --scores scored_regulators.csv \
```

```
--annotations-dir data/annotations \
--top 30 --months 12
```

Step 3: Extract domain signals

```
python signals.py \
--annotations annotations_top30.json \
--output signals.csv \
--events-output events.csv
```

8.2 Integration patterns

- **Dashboard layer** — Visualize weekly signals as timeseries charts per domain and corridor. Highlight active triggers.
- **Scheduled re-solve** — Run the signal extraction weekly. When any trigger flips to active, queue an LP re-solve with adjusted parameters.
- **Scenario planning** — Use monthly aggregation of signals to define stress/relaxed scenarios for quarterly planning cycles.

9. Regulator scoring methodology

The score-regulators step ranks all regulators found in the annotation data using three dimensions:

Dimension	Weight	Method
Frequency	30%	Log-normalized count of updates in the lookback window. Computed from raw data (not LLM). Prevents high-volume regulators from compressing everyone else.
Geo fit	30%	LLM-scored (0–10): how directly does this regulator govern the target region? Regulator name, base URL, and update counts are provided as context.
Client fit	40%	LLM-scored (0–10): how likely does this regulator's activity materially affect the client? Considers industry, geography, and business functions.

Final score = frequency × 0.30 + geo_fit × 0.30 + client_fit × 0.40. Client fit receives the highest weight because it is the most client-specific dimension — two clients in the same geography but different industries will have very different regulator relevance profiles.

10. Limitations and next steps

- **Domain classification is rule-based** — relies on tag and function matching. Annotations with sparse or missing tags may be classified as 'unclassified' (27 rows in current output, 2.5% of total).
- **Corridor inference from metadata** — uses base_url TLD and tag content. Updates from international bodies (FATF, BIS) that affect specific corridors may default to 'global' when the annotation lacks country-level tags.

- **Trigger thresholds need calibration** — the 80th percentile / 4-week window defaults are reasonable starting points but should be tuned against historical supply chain disruption events for the client's specific corridors.
- **Parameter multipliers are illustrative** — the 1.10–1.20× ranges in the mapping table are starting defaults. Actual values should be derived from the client's cost structure and historical regulatory impact data.
- **Temporal coverage** — signals are only as complete as the annotation data. Regulators added to monitoring mid-period will show a step change in their signal history.

Planned enhancements: LLM-assisted domain classification for edge cases, automatic corridor refinement from annotation summaries, and a feedback loop from LP solve results to trigger threshold calibration.